

CSE 112 – Lab #1 Inventory Bins – Array of Structures

Ref: Chapter 11 Programming Challenge #14 (page 674)

Using the main function on the next page, write a program that simulates inventory bins in a warehouse. Each bin holds a number of the same type of parts. The program shall use an array of 10 structures, and 3 functions: a display inventory function, add function, and remove function. The use of a header file for the functions is optional.

The program will create an array of ten (10) structures, initialize the struct members with the data on page 674 (initialization included on next page), and meet with the requirements of the challenge. Users select a bin by number and can add or remove a number of parts.

Input should be validated. The program shall not accept negative numbers to add or remove, a user cannot remove more items than are in the bin, and a bin cannot exceed 30 parts.

Do not use while(true), while(1), break, or continue in the solution.

Grading will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation. The grade penalty is 3 points for each standards violation and 5 points for each missing requirement.

Sample output:

```
Warehouse Inventory
Bin # 1, Part: Valve, Quantity: 10
Bin # 2, Part: Bearing, Quantity: 5
Bin # 3, Part: Bushing, Quantity: 15
Bin # 4, Part: Coupling, Quantity: 21
Bin # 5, Part: Flange, Quantity: 7
Bin # 6, Part: Gear, Quantity: 5
Bin # 7, Part: Gear Housing, Quantity: 5
Bin # 8, Part: Vacuum Gripper, Quantity: 25
Bin # 9, Part: Cable, Quantity: 18
Bin # 10, Part: Rod, Quantity: 12

Enter 0 to quit, or choose a bin Number: 2
Add or Remove Parts (A or R)? A
How many parts to add? 6
Warehouse Inventory
Bin # 1, Part: Valve, Quantity: 10
Bin # 2, Part: Bearing, Quantity: 11
Bin # 3, Part: Bushing, Quantity: 15
Bin # 4, Part: Coupling, Quantity: 21
Bin # 5, Part: Flange, Quantity: 7
Bin # 6, Part: Gear, Quantity: 5
Bin # 7, Part: Gear Housing, Quantity: 5
Bin # 8, Part: Vacuum Gripper, Quantity: 25
Bin # 9, Part: Cable, Quantity: 18
Bin # 10, Part: Rod, Quantity: 12

Enter 0 to quit, or choose a bin Number:
```

Assignments submitted after the due date will incur a **grade penalty of three (3) points per day** after the first day, and **will not be accepted more than one (1) week late**.

// Main function to be used for the program

```

int main()
{
    int binNum = -1;           // Holds the user's menu choice
    string addRem;
    int numParts;

    InventoryBin bins[NUM_BINS] = {           // Array of InventoryBins
        {"Valve", 10},
        {"Bearing", 5},
        {"Bushings", 15},
        {"Coupling", 21},
        {"Flange", 7},
        {"Gear", 5},
        {"Gear Housing", 5},
        {"Vacuum Gripper", 25},
        {"Cable", 18},
        {"Rod", 12}};

    while(binNum != 0)
    {
        binNum = display_inventory(bins);      // call the display function

        if (binNum != 0)
        {
            cout << "Add or Remove Parts (A or R)? ";    // Get user input (A or R)
            cin >> addRem;

            if (addRem == "A")
            {
                add_parts(binNum, bins);                // call the add function
            }
            else if (addRem == "R")
            {
                remove_parts(binNum, bins);              // call the remove function
            }
        } // end of if bin not 0
    } // end of while loop

    return 0;
}

```

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